PSEUDOCODE WRITING

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QUESTIONS

Q1
Student Management System [BL3 - Apply]
10 marks

Q2
Library Book Management [BL3 - Apply]
10 marks

Q3
Employee Salary Calculation [BL3 - Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 - Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 - Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 - Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 - Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 - Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 - Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 - Analyze]
10 marks

10/10 answered

Q1 **Student Management System [BL3 – Apply]** 10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:
- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
CLASS Student
    // Private attributes (Data Hiding & Encapsulation)
    PRIVATE String name
    PRIVATE Integer rollNo
    PRIVATE Array of Float marks
    // Method to accept student details
    PUBLIC METHOD setDetails(sName, sRollNo, sMarks)
        name = sName
        rollNo = sRollNo
        marks = sMarks
    END METHOD
    // Method to calculate average marks
    PUBLIC METHOD calculateAverage()
        IF LENGTH(marks) == 0 THEN
            RETURN 0.0
        END IF
        sum = 0
        FOR EACH mark IN marks DO
            sum = sum + mark
        END FOR

        RETURN sum / LENGTH(marks)
    END METHOD
    // Method to display result (Pass/Fail)
    PUBLIC METHOD displayResult()
        avg = calculateAverage()
        PRINT "Name: " + name
        PRINT "Roll Number: " + rollNo
        PRINT "Average Marks: " + avg
        IF avg >= 50 THEN
            PRINT "Result: PASS"
        ELSE
            PRINT "Result: FAIL"
```

QUESTIONS

Q1
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Shape Area Calculator using Polymorphism [BL3 – Apply]
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5. Vehicle Registration System [BL3 – Apply]
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10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q2 **Library Book Management [BL3 – Apply]** 10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:
- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.
OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
CLASS Book
  // Private attributes (Data Hiding & Encapsulation)
  PRIVATE String title
  PRIVATE String author
  PRIVATE String ISBN
  PRIVATE Boolean isAvailable
  // Constructor / Initialization Method
  PUBLIC METHOD setDetails(bTitle, bAuthor, bISBN)
    title = bTitle
    author = bAuthor
    ISBN = bISBN
    isAvailable = TRUE  // Initially available
  END METHOD
  // Method to issue a book
  PUBLIC METHOD issueBook()
    IF isAvailable == TRUE THEN
      isAvailable = FALSE
      PRINT "Book '" + title + "' issued successfully."
    ELSE
      PRINT "Book '" + title + "' is currently not available."
    END IF
  END METHOD
  // Method to return a book
  PUBLIC METHOD returnBook()
    IF isAvailable == FALSE THEN
```

QUESTIONS

Q1
Student Management System [BL3 - Apply]
10 marks

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Library Book Management [BL3 - Apply]
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10 marks

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Shape Area Calculator using Polymorphism [BL3 - Apply]
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6. Bank Account Hierarchy [BL4 - Analyze]
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7. Hospital Patient Record System [BL4 - Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 - Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 - Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 - Analyze]
10 marks

10/10 answered

Q3 Employee Salary Calculation [BL3 – Apply]

10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: •
Calculate gross salary (basic + HRA + DA)
• Calculate net salary after deductions (PF, tax)
• Display pay slip
Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
CLASS Employee
  // Private attributes (Data Hiding & Encapsulation)
  PRIVATE String empId
  PRIVATE String name
  PRIVATE String designation
  PRIVATE Float basicPay
  // Method to accept employee details
  PUBLIC METHOD setDetails(eId, eName, eDesignation, eBasicPay)
    empId = eId
    name = eName
    designation = eDesignation
    basicPay = eBasicPay
  END METHOD
  // Method to calculate gross salary (basic + HRA + DA)
  PUBLIC METHOD calculateGrossSalary()
    // Assuming HRA = 20% of basic, DA = 10% of basic
    hra = basicPay * 0.20
    da = basicPay * 0.10
    grossSalary = basicPay + hra + da
    RETURN grossSalary
  END METHOD
  // Method to calculate net salary after deductions (PF, Tax)
  PUBLIC METHOD calculateNetSalary()
    grossSalary = calculateGrossSalary()
    // Assuming PF = 12% of basic, Tax = 10% of gross
    pf = basicPay * 0.12
    tax = grossSalary * 0.10
    totalDeductions = pf + tax
    netSalary = grossSalary - totalDeductions
    RETURN netSalary
  END METHOD
  // Method to display pay slip
  PUBLIC METHOD displayPaySlip()
    gross = calculateGrossSalary()
    net = calculateNetSalary()
    PRINT "========== PAY SLIP =========="
    PRINT "Employee ID: " + empId
    PRINT "Name: " + name
    PRINT "Designation: " + designation
    PRINT "Basic Pay: " + basicPay
```

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QUESTIONS

Q1 Student Management System [BL3 - Apply] 10 marks
Q2 Library Book Management [BL3 - Apply] 10 marks
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Q4 Shape Area Calculator using Polymorphism [BL3 - Apply] 10 marks
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Q7 7. Hospital Patient Record System [BL4 - Analyze] 10 marks
Q8 8. E Commerce Product Catalog [BL4 - Analyze] 10 marks
Q9 9. University Staff Hierarchy [BL4 - Analyze] 10 marks
Q10 10. Smart Home Device Controller [BL4 - Analyze] 10 marks

10/10 answered

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply]

10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
// Base Class
CLASS Shape
  // Method to be overridden by derived classes
  PUBLIC METHOD calculateArea()
    RETURN 0.0
  END METHOD
END CLASS
// Derived Class: Circle
CLASS Circle EXTENDS Shape
  PRIVATE Float radius
  PUBLIC METHOD setRadius(r)
    radius = r
  END METHOD
  // Overriding calculateArea method
  OVERRIDE PUBLIC METHOD calculateArea()
    RETURN 3.14159 * radius * radius
  END METHOD
END CLASS
// Derived Class: Rectangle
CLASS Rectangle EXTENDS Shape
  PRIVATE Float length
  PRIVATE Float breadth
  PUBLIC METHOD setDimensions(l, b)
    length = l
    breadth = b
  END METHOD
  // Overriding calculateArea method
  OVERRIDE PUBLIC METHOD calculateArea()
    RETURN length * breadth
  END METHOD
END CLASS
// Derived Class: Triangle
CLASS Triangle EXTENDS Shape
  PRIVATE Float base
  PRIVATE Float height
  PUBLIC METHOD setDimensions(b, h)
    base = b
    height = h
  END METHOD
```

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QUESTIONS

- **Q1** Student Management System [BL3 – Apply] 10 marks
- **Q2** Library Book Management [BL3 – Apply] 10 marks
- **Q3** Employee Salary Calculation [BL3 – Apply] 10 marks
- **Q4** Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks
- **Q5** 5. Vehicle Registration System [BL3 – Apply] 10 marks
- **Q6** 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks
- **Q7** 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
- **Q8** 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks
- **Q9** 9. University Staff Hierarchy [BL4 – Analyze] 10 marks
- **Q10** 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

10/10 answered

Q5 5. Vehicle Registration System [BL3 – Apply]

10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
// Base Class
CLASS Vehicle
   // Protected attributes to allow direct access in subclasses
   PROTECTED String regNo
   PROTECTED String owner
   PROTECTED String fuelType
   // Method to set vehicle details
   PUBLIC METHOD setVehicleDetails(rNo, vOwner, fType)
      regNo = rNo
      owner = vOwner
      fuelType = fType
   END METHOD
   // Method to display base details
   PUBLIC METHOD displayDetails()
      PRINT "Registration Number: " + regNo
      PRINT "Owner Name: " + owner
      PRINT "Fuel Type: " + fuelType
   END METHOD
END CLASS
// Subclass: TwoWheeler
CLASS TwoWheeler EXTENDS Vehicle
   PRIVATE Integer engineCapacity // in CC
   // Method to set TwoWheeler details
   PUBLIC METHOD setTwoWheelerDetails(rNo, vOwner, fType, eCapacity)
      CALL setVehicleDetails(rNo, vOwner, fType)
      engineCapacity = eCapacity
   END METHOD
   // Overriding displayDetails method
   OVERRIDE PUBLIC METHOD displayDetails()
      PRINT "=== Two-Wheeler Details ==="
      CALL SUPER.displayDetails()
      PRINT "Engine Capacity: " + engineCapacity + " CC"
   END METHOD
END CLASS
// Subclass: FourWheeler
CLASS FourWheeler EXTENDS Vehicle
   PRIVATE Integer seatingCapacity
   PRIVATE Boolean hasAC
   // Method to set FourWheeler details
   PUBLIC METHOD setFourWheelerDetails(rNo, vOwner, fType, sCapacity, acStatus)
      CALL setVehicleDetails(rNo, vOwner, fType)
```

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7. Hospital Patient Record System [BL4 – Analyze]
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Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

Q6 6. Bank Account Hierarchy [BL4 – Analyze]

10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
// Base Class
CLASS BankAccount
    PROTECTED String accNo
    PROTECTED String holder
    PROTECTED Float balance
    PUBLIC METHOD setAccountDetails(accountNumber, accountHolder, initialBalance)
        accNo = accountNumber
        holder = accountHolder
        balance = initialBalance
    END METHOD
    PUBLIC METHOD deposit(amount)
        IF amount > 0 THEN
            balance = balance + amount
            PRINT "Deposited: " + amount
        END IF
    END METHOD
    PUBLIC METHOD withdraw(amount)
        IF amount > 0 AND amount <= balance THEN
            balance = balance - amount
            PRINT "Withdrawn: " + amount
        ELSE
            PRINT "Insufficient funds!"
        END IF
    END METHOD
    PUBLIC METHOD displayBalance()
        PRINT "Account Number: " + accNo
        PRINT "Holder Name: " + holder
        PRINT "Current Balance: " + balance
    END METHOD
END CLASS
// Subclass: SavingsAccount
CLASS SavingsAccount EXTENDS BankAccount
    PRIVATE Float interestRate
    PUBLIC METHOD setSavingsDetails(accountNumber, accountHolder, initialBalance, rate)
        CALL setAccountDetails(accountNumber, accountHolder, initialBalance)
        interestRate = rate
    END METHOD
    PUBLIC METHOD calculateInterest()
        interest = balance * (interestRate / 100)
        balance = balance + interest
```

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Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
Q8 8. E Commerce Product Catalog [BL4 – Analyze] 10 marks
Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks
Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

10/10 answered

Q7 7. Hospital Patient Record System [BL4 – Analyze]

10 marks

Design a Patient base class and analyze how it extends into:
• InPatient — with ward number, admission date, and daily charges
• OutPatient — with appointment date and consultation fee
Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
// Base Class
CLASS Patient
    PROTECTED String patientId
    PROTECTED String name
    PROTECTED Integer age
    // Private attributes for Encapsulation of sensitive medical data
    PRIVATE String diagnosis
    PRIVATE String prescription
    PUBLIC METHOD setPatientDetails(pId, pName, pAge)
        patientId = pId
        name = pName
        age = pAge
    END METHOD
    // Controlled accessors for sensitive medical data
    PUBLIC METHOD setMedicalData(pDiagnosis, pPrescription)
        diagnosis = pDiagnosis
        prescription = pPrescription
    END METHOD
    PUBLIC METHOD getDiagnosis()
        RETURN diagnosis
    END METHOD
    PUBLIC METHOD getPrescription()
        RETURN prescription
    END METHOD
    PUBLIC METHOD generateBill()
        RETURN 0.0
    END METHOD
END CLASS
// Subclass: InPatient
CLASS InPatient EXTENDS Patient
    PRIVATE String wardNumber
    PRIVATE String admissionDate
    PRIVATE Float dailyCharges
    PRIVATE Integer numberOfDays
    PUBLIC METHOD setInPatientDetails(pId, pName, pAge, ward, admDate, dailyFee, days)
        CALL setPatientDetails(pId, pName, pAge)
        wardNumber = ward
        admissionDate = admDate
        dailyCharges = dailyFee
        numberOfDays = days
    END METHOD
    OVERRIDE PUBLIC METHOD generateBill()
```

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Q5
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Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

Q8 **8. E-Commerce Product Catalog [BL4 – Analyze]**

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
// Base Class
CLASS Product
    PROTECTED String productId
    PROTECTED String name
    PROTECTED Float price
    PUBLIC METHOD setProductDetails(pId, pName, pPrice)
        productId = pId
        name = pName
        price = pPrice
    END METHOD
    PUBLIC METHOD getPrice()
        RETURN price
    END METHOD
    PUBLIC METHOD applyDiscount()
        // Default discount is 0%
        RETURN price
    END METHOD
END CLASS
// Subclass: Electronics
CLASS Electronics EXTENDS Product
    PRIVATE Integer warrantyMonths
    PRIVATE String brand
    PRIVATE Integer voltage
    PUBLIC METHOD setElectronicsDetails(pId, pName, pPrice, warranty, pBrand, pVoltage)
        CALL setProductDetails(pId, pName, pPrice)
        warrantyMonths = warranty
        brand = pBrand
        voltage = pVoltage
    END METHOD
    OVERRIDE PUBLIC METHOD applyDiscount()
        // 10% discount on Electronics
        discountedPrice = price - (price * 0.10)
        RETURN discountedPrice
    END METHOD
END CLASS
// Subclass: Clothing
CLASS Clothing EXTENDS Product
    PRIVATE String size
    PRIVATE String fabric
    PRIVATE String gender
```

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Employee Salary Calculation [BL3 - Apply]
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Q5
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Q7
7. Hospital Patient Record System [BL4 - Analyze]
10 marks

Q8
8. E Commerce Product Catalog [BL4 - Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 - Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 - Analyze]
10 marks

10/10 answered

Q9 **9. University Staff Hierarchy [BL4 – Analyze]**

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
// Base Class
CLASS Staff
    PROTECTED String staffId
    PROTECTED String name
    PROTECTED Float baseSalary
    PUBLIC METHOD setStaffDetails(sId, sName, salary)
        staffId = sId
        name = sName
        baseSalary = salary
    END METHOD
    PUBLIC METHOD calculatePay()
        RETURN baseSalary
    END METHOD
    PUBLIC METHOD displayDetails()
        PRINT "Staff ID: " + staffId
        PRINT "Name: " + name
        PRINT "Base Salary: " + baseSalary
    END METHOD
END CLASS
// Subclass: Professor
CLASS Professor EXTENDS Staff
    PRIVATE String department
    PRIVATE Float researchAllowance
    PUBLIC METHOD setProfessorDetails(sId, sName, salary, dept, allowance)
        CALL setStaffDetails(sId, sName, salary)
        department = dept
        researchAllowance = allowance
    END METHOD
    OVERRIDE PUBLIC METHOD calculatePay()
        // Total pay includes base salary + research allowance
        totalPay = baseSalary + researchAllowance
```

Save Answer

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10 marks

Q6
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10 marks

Q7
7. Hospital Patient Record System [BL4 - Analyze]
10 marks

Q8
8. E Commerce Product Catalog [BL4 - Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 - Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 - Analyze]
10 marks

10/10 answered

Q10 **10. Smart Home Device Controller [BL4 – Analyze]** 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
// Base Interface / Abstract Base Class
CLASS SmartDevice
    PROTECTED String deviceId
    PROTECTED String deviceName
    PROTECTED Boolean status
    PUBLIC METHOD setDeviceDetails(dId, dName)
        deviceId = dId
        deviceName = dName
        status = FALSE
    END METHOD
    PUBLIC METHOD turnOn()
        status = TRUE
        PRINT deviceName + " is now ON"
    END METHOD
    PUBLIC METHOD turnOff()
        status = FALSE
        PRINT deviceName + " is now OFF"
    END METHOD
    PUBLIC METHOD getStatus()
        RETURN status
    END METHOD
END CLASS
// Derived Class: SmartLight
CLASS SmartLight EXTENDS SmartDevice
    PRIVATE Integer brightness
    PRIVATE String color
    PUBLIC METHOD setLightDetails(dId, dName)
        CALL setDeviceDetails(dId, dName)
        brightness = 100
        color = "White"
    END METHOD
    PUBLIC METHOD setBrightness(level)
        IF status == TRUE THEN
            brightness = level
            PRINT deviceName + " brightness set to " + brightness + "%"
        ELSE
            PRINT "Cannot adjust brightness. " + deviceName + " is OFF"
        END IF
    END METHOD
```